



3.4 - Tricuspid Regurgitation SOAP

1) Tri-pyramids:

Tricuspid valve

2) Water backflowing out of mouth:

Tricuspid Regurgitation = backflow of blood from RV to RA during systole

Symbols

3) “Physio” water bottle:

“Physiologic” **TR** is present in ~70% of normal adults

4) Torn #1 saddle with tri-pyramids:

Primary TR (caused by conditions directly affecting the tricuspid valve) is rare

5) #2 saddle:

Secondary TR (regurgitation in the setting of an anatomically normal tricuspid valve) = most common type of pathologic TR

6) Dilated camel leash:

Secondary **TR** is caused by conditions that result in **dilation of tricuspid annulus**

Symbols

7) Happy camel:

Most patients with mild-moderate **TR** are **asymptomatic**

8) Floppy heart balloon:

Patients with severe **TR** most commonly present with symptoms of **right-sided HF** (e.g., peripheral edema, hepatomegaly, ascites) (**Chief Complaint**)

9) Tight red chest laces:

Secondary/Functional **TR** can be caused by **pulmonary HTN (e.g., 2/2 left-sided HF, cor pulmonale, mitral valve disease)**, left-to-right shunts, & RV cardiomyopathy (**PMHx**)

10) Left-to-right arrow:

Secondary/Functional **TR** can be caused by pulmonary HTN, **left-to-right shunts (e.g., VSD, ASD)**, & RV cardiomyopathy (**PMHx**)

11) Dilated heart bag:

Secondary/Functional **TR** can be caused by pulmonary HTN, left-to-right shunts (e.g., VSD, ASD), & **RV cardiomyopathy (e.g., dilated cardiomyopathy, ischemic heart disease) (PMHx)**

12) Spice canisters:

Carcinoid syndrome can cause primary **TR** → due to plaque-like deposits of fibrous tissue on the tricuspid valve (**PMHx**)

13) Silver chest pin with wires:

Damage to tricuspid valve from placement of **implanted pacemaker or ICD leads** can cause primary **TR** (**PMHx**)

14) Heart lantern:

Infective endocarditis 2/2 IV drug use can cause tricuspid vegetations & primary **TR** (**Social Hx**)

15) Ivy-strapped drug bag:

Infective endocarditis 2/2 **IV drug use** can cause tricuspid vegetations & primary **TR** (**Social Hx**)

16) Blue JVP jugs:

Severe **TR** is associated with signs of right-sided HF (e.g., **elevated JVP**, pitting pretibial edema, hepatojugular reflux) (**Physical Exam**)

17) Baggy pant legs:

Severe **TR** is associated with signs of right-sided HF (e.g., elevated JVP, **pitting pretibial edema**, hepatojugular reflux) (**Physical Exam**)

18) Dog with liver-shaped spot knocking over blue jug:

Severe **TR** is associated with signs of right-sided HF (e.g., elevated JVP, pitting pretibial edema, **hepatojugular reflux**)(**Physical Exam**)

19) Pulsatile liver dog:

Severe **TR** may cause a **pulsatile liver** on abdominal palpation (**Physical Exam**)

20) Murmur lines:

Classic **TR** **murmur** = blowing holosystolic murmur best heard at the left lower sternal border(**Cardiac Auscultation**)

21) Hologram:

Classic **TR** murmur = blowing **holosystolic** murmur best heard at the left lower sternal border(**Cardiac Auscultation**)

22) Customer gasping & raising leg:

Classic murmur of **TR** increases with maneuvers that increase venous return (e.g., inspiration, leg raise) (**Cardiac Auscultation**)

23) Heart-shaped megaphone:

TTE = imaging modality of choice for diagnosis of **TR** (**Echo**)

24) DDx clipboard:

For further workup of Tricuspid Regurgitation, see Chronic **HF** SOAP

25) Ankhs with big loops:

In patients with severe **TR** and signs of right-sided HF, **loop diuretics** (e.g., furosemide) can improve symptoms **(1)**

26) Repairing UNDERBODY of car:

In patients with secondary **TR**, **treating the underlying etiology** can improve or reverse valvular dysfunction **(2)**

27) Scalpel:

Tricuspid valve surgery (repair / replacement) is recommended for most patients with severe, symptomatic disease **(3)**